

El Mahdi Moubarak

Software & AI Engineer · Full-Stack, LLM Systems, Agents

Lille, France

elmahdi@moubarak.dev · moubarak.dev · linkedin.com/in/el-mahdi-moubarak · github.com/emoubarak

SUMMARY

Full-stack engineer with 6 years shipping production software in cybersecurity, healthcare and fintech, most of the time as the only engineer on the product. Since 2026 I build AI-native: a browser agent that tests my SaaS as real users with the database as judge, LLM features whose models are picked by attacking them, and Claude Code as my main engineering environment.

EXPERIENCE

Co-Founder & CTO · **Piktechs** , France

Mar 2026 – Present

- **Sole engineer:** took a two-week prototype to a **complete production B2B SaaS** for event lead capture, with Claude Code as the main engineering environment
- **Agentic QA with jev**, Browser Use's browser agent: it signs in to the test app as each account state (free, Pro, lapsed, team member) and follows a plain-English goal, then a **SQL query, not the agent, gives the verdict. 16 standing checks** on plan gates and permissions; its before/after screenshots on closed tickets surfaced **3 follow-up bugs**. Open-sourced as [jev-check](#)
- Shipped an **AI business-card scanner**: vision LLM with strict JSON output, per-user quota and on-device OCR as last resort; chose the cross-provider fallback by **prompt-injection testing (2 of 4 candidate models obeyed an injected card)**
- Built **live Stripe billing** (checkout, portal, webhooks, per-seat plans, reconciliation) and multi-tenant workspaces secured by **Postgres row-level security**; billing and permission code is **reviewed by a different model** from the one that wrote it
- Shipped the **Android app**, [live on Google Play](#), and the iOS build from one codebase (Capacitor); unit and Playwright tests gate CI/CD across separate test and production environments

Stack React · TypeScript · Vite · Tailwind · Supabase (Postgres, RLS, Edge Functions) · Stripe · Capacitor · OpenRouter · Resend · Playwright, Vitest · jev (Browser Use) · GitHub Actions · Cloudflare Pages · Claude Code

Founder & AI / Full-Stack Engineer · **EMM** , Lille, France

Mar 2026 – Present

- Independent engineering practice: **LLM and OCR pipelines**, **AI agents** with custom skills and MCP, and browser automation with jev for back-offices that have no API, built for SMEs and agencies
- **AI-native workflow:** a versioned Claude Code workspace shared across two machines, where repeated corrections become hooks and skills, and a weekly pass mines my own sessions for what I keep correcting ([write-up](#))
- Designed and delivered **corporate AI training** for DRIVECO (EV charging operator): two FR/EN sessions built on the company's real workflows, with a [public testimonial](#) from its Chief People Officer
- Also CTO of **KSUR Services** (web and automation agency, France and Morocco): own the technical stack and delivery

Stack TypeScript · Python · Next.js · Node.js · Supabase · Docker · OpenAI, Claude, Gemini APIs · MCP · Browser Use · Claude Code

Full-Stack Engineer · **Board of Cyber** , Paris (remote)

Mar 2023 – Feb 2026

- Core engineer on *Security Rating*®, a SaaS cyber-rating platform used by **200+ organizations**, across the client platform, back office and manager office
- Built a Python **vulnerability-scanning probe** on nmap that automated attack-surface data collection for the ratings

- Added test coverage (Spectator), optimized **CI/CD pipelines** and contributed to the Angular migration and major library upgrades

Stack Angular · TypeScript · Python · nmap · Spectator · Docker · CI/CD

Full-Stack & AI Engineer · **ALTAO Santé** , Lille

Sep 2021 – Sep 2022

- **Sole engineer** on a greenfield medical platform correcting hospital discharge summaries, replacing a fragmented multi-tool workflow
- Processed **5,000+ documents** through the platform, saving hospitals **€700,000**
- Integrated the Data Science team's **ML prediction models** and rule engines into the production workflow
- Ran requirement sessions with doctors, wrote the specs, deployed and monitored the AWS infrastructure

Stack Vue.js · Django · Python · PostgreSQL · AWS (EC2, Route53, Amplify)

Mobile Software Engineer · **Deuspi** , Lille

Mar 2021 – Jun 2021

- **Sole mobile engineer** on the first version of a React Native grocery delivery app, from designer wireframes to production

Stack React Native · JavaScript · REST APIs

Full-Stack Engineer · **Worldline** , Seclin

Apr 2019 – Aug 2019 (internship)

- Built an internal R&D tool to create, store and replay payment test sets over MQTT, improving the team's development speed by **70%**

Stack Vue.js · Spring Boot · Java · MQTT · Docker

OPEN SOURCE & PROJECTS

- **jev-check** : open-source (MIT) agent skill for Claude Code, Codex and OpenClaw: **browser-agent QA** on a staging app where a **SQL query, not the agent**, gives the verdict (Python, CDP)
- **jev-administrator**: self-hosted orchestrator for browser agents (private): persistent logged-in browser profiles, one isolated worker per task, **live human hand-off** over WebRTC for logins and 2FA, back-office with a task timeline (Python, Podman, CDP)
- **Polymarket Lab** : six weeks of trading-edge research: **out-of-sample backtests**, **~12,000 paper trades** across 60 concurrent runners, Python engine and Go **market maker**. Verdict: **no tradable edge**
- **PDFold** : PDF to structured Markdown in **11 languages**: **multi-pass OCR + Gemini pipeline** with failover, Stripe billing (Next.js, Supabase)
- **AI Visibility Tracker** : measures how ChatGPT, Perplexity, Gemini and Google AI Overviews **mention a brand**, **sampling each prompt several times** per engine; callable over **MCP** (TypeScript)
- **Writing** : **how I work with AI**; case studies on [the AI card scanner](#), [jev-check](#) and [Piktechs](#)

SKILLS

Programming

TypeScript, JavaScript Python Go SQL Java

AI & LLM

LLM APIs (OpenAI, Anthropic Claude, Gemini, OpenRouter) AI agents, MCP, Claude Code

Browser agents & agentic QA (jev, Browser Use) LLM evaluation & prompt-injection testing

OCR & vision pipelines RAG Local LLMs (llama.cpp, ExLlamaV2)

Web, Data & Cloud

React, Next.js, Node.js Angular, Vue.js Django React Native, Capacitor

PostgreSQL, Supabase Stripe AWS, Cloudflare Docker, CI/CD

Languages

French (native) English (C1, TOEIC) Spanish (professional)

EDUCATION

IMT Nord Europe · Engineering Degree (Master's level), CS, Networks & Telecoms

Lille, 2017 – 2022